Chapter 7 • Coordinate Geometry

Solved Review Exercise 7

MCQs

1. The slope–intercept form is:
a) $y=m(x+c)$
b) $y-y_1=m(x-x_1)$
c) $y=c+mx$
d) $ax+by+c=0$
Ans: c
$y=c+mx$ is $y=mx+c$.

2. Gradients of two parallel lines are:
a) equal
b) zero
c) negative reciprocals
d) always undefined
Ans: a
Parallel non-vertical lines have equal gradients.

3. If the product of two gradients is −1, the lines are:
a) parallel
b) perpendicular
c) collinear
d) coincident
Ans: b
This is the perpendicular-gradient test.

4. Distance between P(1,2) and Q(4,6) is:
a) 5
b) 6
c) $\sqrt{13}$
d) 4
Ans: a
$\sqrt{(3^2+4^2)}=5$.

5. Midpoint of (−2,4) and (6,−2) is:
a) (4,2)
b) (2,1)
c) (1,1)
d) (0,0)
Ans: b
Averaging coordinates gives (2,1).

6. A line through (1,2) and (4,5) is:
a) $y=x+1$
b) $y=2x+3$
c) $y=3x-2$
d) $y=x+2$
Ans: a
Slope=1 and intercept=1.

7. The point–slope form is:
a) $y=m(x+c)$
b) $y-y_1=m(x-x_1)$
c) $y=c+mx$
d) $ax+by+c=0$
Ans: b
It uses one point and the gradient.

8. $2x+3y-6=0$ in slope–intercept form is:
a) $y=\frac{-2x}{3}+2$
b) $y=\frac{2x}{3}-2$
c) $y=\frac{-3x}{2}+2$
d) $y=\frac{3x}{2}+2$
Ans: a
$3y=-2x+6$.

9. The symmetric form is:

studysapiens.com

a) x/a+y/b=1
b) a three-variable ratio
c) $\frac{x-x_1}{\cos}\alpha=\frac{y-y_1}{\sin}\alpha=r$
d) $y-y_1=m(x-x_1)$
Ans: c
It uses direction cosines and a parameter r.

10. The normal form is:
a) y=mx+c
b) x/a+y/b=1
c) $\frac{x-x_1}{\cos}\alpha=\frac{y-y_1}{\sin}\alpha$
d) xcosα+ysinα=p
Ans: d
p is the perpendicular distance from the origin.

2. Find the distance between A(2,3) and B(7,8).
d = $\sqrt{(5^2+5^2)}$
= $5\sqrt{2}$ units.

3. Find the midpoint of (4,−2) and (−6,3).
M=$(-1,\frac{1}{2})$.

4. Find the gradient through (1,2) and (4,6).
m = $\frac{6-2}{4-1}$
= $\frac{4}{3}$

5. Find y=mx+c through (3,7) and (5,11).
m=$\frac{11-7}{5-3}$=2. Using (3,7): 7=2(3)+c ⇒ c=1.
y=2x+1.

6. Two lines are parallel and one has gradient $\frac{2}{3}$ Find the other gradient.

The other gradient is also $\frac{2}{3}$

7. Find the straight-line distance from A(12,5) to B(8,−4).
d = $\sqrt{[(-4)^2+(-9)^2]}$
= $\sqrt{97}\approx9.85$ units.

8. A path starts at (2,3) and ends at (10,7). Find the midpoint.
M=(6,5).

9. A drone flies from (2,3) to (10,15). Find gradient and distance.
m = $\frac{12}{8}$
= $\frac{3}{2}$; d
= $\sqrt{(8^2+12^2)}$
= $4\sqrt{13}\approx14.42$ units.

10. A line has gradient −3 and y-intercept 2. Write all requested forms and address the supplied points.

(a) Stated line: y=−3x+2. Its x-intercept is $\frac{2}{3}$

(b) Literal point–slope using (1,2): y−2=−3(x−1). This describes y=−3x+5, not the stated line.

(c) Literal two-point form: $\frac{y-2}{x-1}=\frac{-7-2}{4-1}=-3$; again y=−3x+5.

(d) For the stated line: $\frac{x}{\frac{2}{3}}\frac{+y}{2}=1$.

(e) $\frac{x}{\frac{1}{\sqrt{10}}}=\frac{y-2}{\frac{-3}{\sqrt{10}}}$
= r.

(f) Normal form for 3x+y=2: $(3/\sqrt{10})x+(1/\sqrt{10})y=\frac{2}{\sqrt{10}}$

studysapiens.com